

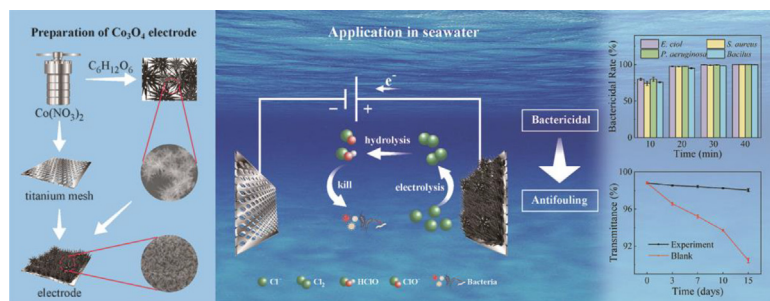


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journal homepage: www.elsevier.com/locate/jcisMorphology regulation and application of nano cobalt oxide (Co_3O_4) electrocatalysts for chlorine evolution toward marine anti-biofoulingHaomin Wang^{a,b,c}, Jiang Li^e, Dun Zhang^{b,c,d}, Guoqing Wang^{a,*}, Peng Wang^{b,c,d,*}^a State Key Laboratory of Marine Resource Utilization in South China Sea, School of Materials Science and Engineering, Hainan University, Haikou 570228, China^b Open Studio for Marine Corrosion and Protection, Pilot National Laboratory for Marine Science and Technology (Qingdao), 168 Wenhai Middle Road, Qingdao 266237, China^c Key Laboratory of Marine Environmental Corrosion and Biofouling, Institute of Oceanology, Chinese Academy of Sciences, 7 Nanhai Road, Qingdao 266071, China^d Center for Ocean Mega-Science, Chinese Academy of Sciences, 7 Nanhai Road, Qingdao 266071, China^e High-Performance Materials Institute, Florida State University, Department of Industrial and Manufacturing Engineering, FAMU-FSU College of Engineering, 2525 Pottsdamer Street, Tallahassee, FL 32310, USA

GRAPHICAL ABSTRACT



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ABSTRACT

For the optical equipment used in the marine environment, the quality and accuracy of the tests are easily affected by biological fouling. Hence, good anti-fouling performance is crucial for preserving the optical equipment. A new electrochemical chlorination anti-fouling system has been proposed for years and is suitable for underwater optical windows. In this paper, nano Co_3O_4 electrocatalysts with different morphologies were prepared through the hydrothermal method to study the structure–activity relationship between their electrochemical properties and morphologies. The nanourchin Co_3O_4 shows the best electrochemical properties resulting from its large surface area. The titanium mesh electrode with nanourchin Co_3O_4 was used to evaluate the anti-fouling performance. In comparison to commercial dimensionally stable anode electrodes, the developed Co_3O_4 /titanium mesh electrode has higher catalytic performance, better selectivity, and lower cost. The bactericidal rate can reach 95% in 20 mins and it can effectively inhibit the sedimentation of microorganisms on the glass surface. Therefore, it is a promising strategy to use the nanourchin Co_3O_4 as an electrocatalyst to protect the underwater optical windows.

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1. Introduction

Underwater detection is a very important method for understanding the marine ecosystem and optical monitoring is the most effective approach to obtaining the needed information [1–3]. However, the optical equipment is easily affected by fouling organisms when it is used in a marine environment that has more than

4000 kinds of organisms [4,5]. The microorganisms tend to settle on the out surface of the equipment immersed in seawater to form a conditioned film, and further, a large-scale biofouling layer [6], which will seriously affect the accuracy and service life of optical monitors [7,8]. Frequent manual maintenance is required to keep the performance of the equipment but is expensive and time-consuming [9]. Therefore, it is necessary to find a suitable and efficient way to protect the equipment from bio-fouling in the ocean.

In marine environment, many strategies are used to inhibit the bio-fouling, such as mechanical cleaning [10], rough microstructure [11,12], UV sterilization [13,14], antifouling coatings [15–19], and electrochemical chlorination [20–22]. However, not all antifouling technologies are suitable for optical windows due to their particularity. Mechanical cleaning with scrapers to remove the fouling organisms may damage the optical window surface. Surface rough topographies are applicable to the antifouling of optical windows, but their long-term stability, light transmission efficiency, and surface adhesion in seawater still need further study and optimization [23]. UV antibacterial method is not suitable for optical windows because of its high energy consumption, high cost, and low underwater efficiency. Most antifouling coatings are organic polymers with poor light transmittance which limits their application. Moreover, most antifouling coatings containing copper and organotin, which have a negative impact on the marine ecological environment, have been strictly restricted [24]. Compared to the above methods, electrochemical chlorination has the highest potential for protecting underwater optical windows. The electrocatalyst can directly electrolyze chlorine in seawater to produce Cl_2 , and the chlorine in seawater is infinite. At the same time, the production of active chlorine can be flexibly regulated and will not cause harm to the marine environment [25,26]. In addition, the efficiency of the electrocatalyst is high, and the transmittance of the optical window itself will not be affected because there is no direct contact [27].

In recent years, electrochemical water splitting has been rapidly developing as an environmentally friendly and sustainable strategy [28,29]. Co_3O_4 has attracted much attention as electrocatalysts of the chlorine evolution reaction (CER) due to its high performance and low cost. Heonjin Ha et al. reported that Co_3O_4 showed comparable CER performance to that of commercial RuO_2 Under neutral pH [30]. However, its performance in alkaline seawater is still not clear. Therefore, it is valuable to understand its CER performance in alkaline seawater. In addition, Zhu and Ren found that Co_3O_4 with micro-nano structures had significantly better catalytic efficiency [31,32]. Therefore, a systematic study of the effect of different Co_3O_4 microstructures on the CER performance is possible to achieve a more efficient catalyst.

In this research, Co_3O_4 powders with different morphologies were prepared through the hydrothermal method. By studying the effect of different morphologies on the electrocatalytic performance, a novel Co_3O_4 /titanium mesh (TM) electrode was prepared and its electrocatalytic activity, stability, and selectivity of CER were studied. Finally, the sterilization and antifouling performance of the Co_3O_4 /TM electrode were verified with different bacteria and methods. Therefore, this new CER electrocatalyst is a good candidate for the antifouling treatment of optical windows in seawater.

2. Experimental

2.1. Materials

The TM ($200 \times 1000 \times 0.1$ mm, 99.999 %) was purchased from Chengshuo Metal material Co. Ltd. Cobalt nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), urea ($\text{CH}_4\text{N}_2\text{O}$), ammonium fluoride (NH_4F), glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), sodium chloride (NaCl), and sodium sulfate (Na_2SO_4)

were provided by Shanghai Aladdin Biochemical Technology Co. Ltd. Hexamethylenetetramine ($\text{C}_6\text{H}_{12}\text{N}_4$) and anhydrous ethanol ($\text{C}_2\text{H}_6\text{O}$) were purchased from Sinopharm Chemical Reagent Co. Ltd. The DPD (N, N-diethyl-p-phenylenediamine) Total Chlorine Test Kit was provided by Guangdong Huankai Microbial Sci & Tech. Co. Ltd. Dimensionally stable anode (DSA) electrodes were purchased from Baoji Jinte Technology Materials Co. Ltd. (Ti supported RuO_2 and IrO_2). All chemicals were analytical grade and used directly without further purification.

2.2. Co_3O_4 catalyst and TM electrode preparation

The synthesis flow of Co_3O_4 powder with different morphologies is shown in Fig. 1. Before the hydrothermal reaction, 5 mmol $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and 10 mmol $\text{CH}_4\text{N}_2\text{O}$ were added to 70 mL deionized water, stirred at room temperature for 30 mins to obtain a pink clear solution, and then transferred to 100 mL stainless steel hydrothermal reactor. The reactor was kept at 130°C for 24 h. After cooling down to room temperature, the precursor samples were centrifuged, collected, washed with deionized water and anhydrous ethanol three times respectively, and then dried in a vacuum oven at 60°C for 8 h. Finally, the dried precursor sample was annealed in a muffle furnace at 300°C in the air for 2 h with a heating rate of $1^\circ\text{C} / \text{min}$. The nanosheet powder was directly prepared under the above conditions. The nanorod powder and nanourchin powder were obtained with the extra addition of 10 mmol NH_4F and 6 mmol $\text{C}_6\text{H}_{12}\text{O}_6$, respectively. For nanoflower powder, 1 mmol $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 5 mmol $\text{C}_6\text{H}_{12}\text{N}_4$, and $\text{C}_6\text{H}_{12}\text{O}_6$ with a mass of 1/4 of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ were mixed and dissolved in a solution with a volume ratio of 2:1 of deionized water and anhydrous ethanol. And the hydrothermal reaction temperature is 110°C and the time is 12 h. For Co_3O_4 /TM electrode, TM ($2\text{ cm} \times 4\text{ cm}$) was washed in acetone with ultrasonic for 15 mins, treated in oxalic acid (10 %) at 70°C for 1 h, and then stored in anhydrous ethanol for use. The TM electrode is directly prepared through the hydrothermal reaction when the TM is placed in the reaction kettle of nanourchin powder.

2.3. Catalyst characterization

The morphologies of nanosheet, nanorod, nanourchin, nanoflower, and TM samples were observed by field emission scanning electron microscopy (FE-SEM, Hitachi Regulus 8100, Japan) under the accelerated voltage of 10 kV. The crystal structure of the sample was analyzed through the X-ray diffractometer (XRD, Rigaku, Japan) with Cu K-alpha radiation ($\lambda = 0.15406\text{ nm}$) at the scanning speed of $5^\circ/\text{min}$ in the 2θ range from 10° to 90° . The Raman spectra of four kinds of samples were measured by micro confocal Raman spectroscopy (Renishaw MZ20-FC) with a laser wavelength of 532 nm. The N_2 adsorption-desorption curve was recorded by automatic specific surface area and porosity analyzer (Micromeritics, ASAP2460) at -195.85°C . The specific surface area and pore size distribution of the samples were calculated by Brunauer-Emmett-Teller (BET) and Barrett-Joyner-Halenda (BJH) methods, respectively [33].

2.4. Electrochemical measurements

The electrochemical tests were carried out by CHI760c electrochemical workstation at room temperature in an electrolytic cell with a standard three-electrode mode. A saturated calomel electrode was used as the reference electrode and a piece of the platinum net was used as the counter electrode. The Co_3O_4 /TM and commercial DSA were used as the working electrode, respectively. The electrocatalytic activity of all samples in the range of 0–1.6 V was tested in 0.6 M NaCl solution with a scanning rate of 10 mV/s by the linear sweep voltammetry (LSV). The test frequency of

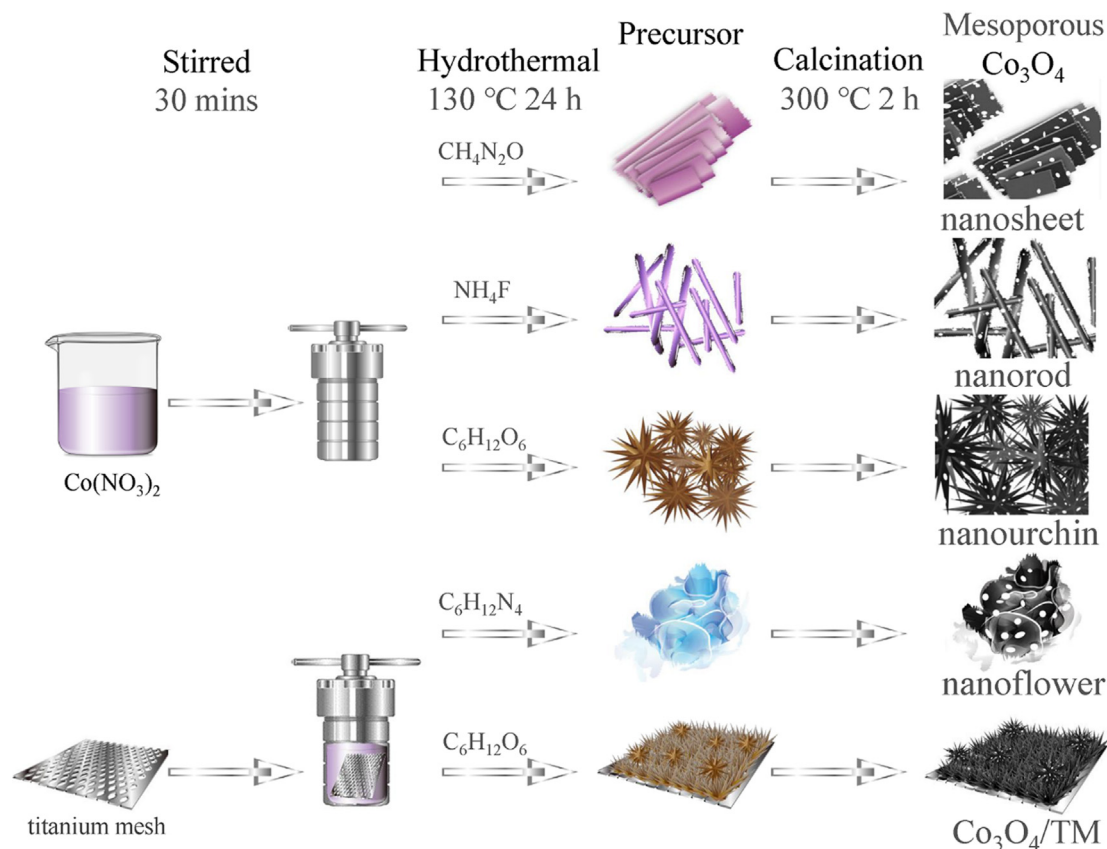


Fig. 1. The synthesis of Co_3O_4 nanopowders with different morphologies (nanosheet, nanorod, nanourchin, and nanoflower) and $\text{Co}_3\text{O}_4/\text{TM}$ electrodes by the hydrothermal method.

electrochemical impedance spectroscopy (EIS) was from 100 kHz to 0.01 Hz and measurements were carried out at 1.2 V. The stability of the DSA electrode and $\text{Co}_3\text{O}_4/\text{TM}$ electrode was evaluated by chronopotentiometry ($5\text{ mA}/\text{cm}^2$ and $10\text{ mA}/\text{cm}^2$). The double-layer capacitance (C_{dl}) of catalyst powder was measured by the cyclic voltammetry (CV, scanning rate, $10\text{--}45\text{ mV}/\text{s}$, $5\text{ mV}/\text{s}$ intervals; potential, $0.4\text{--}0.5\text{ V}$).

2.5. Active chlorine concentration

In 60 mL 0.6 M NaCl solution, the electrolyte containing different concentrations of active chlorine were obtained by chronopotentiometry for 180 s at different current densities (1, 5, 10, 15, $20\text{ mA}/\text{cm}^2$). The electrolyte containing active chlorine was oxidized to purple by DPD active chlorine kit, and the absorbance was measured three times by a porous microplate reader (M1000pro) at 552 nm. The concentration of active chlorine and Faradaic efficiency were calculated by the DPD colorimetric method.

2.6. Evaluation of bactericidal performance

Escherichia coli (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) are commonly used laboratory model bacteria, *Pseudomonas aeruginosa* (*P. aeruginosa*) and *Bacillus* are common marine fouling bacteria. Gram-negative bacteria (*E. coli*, *P. aeruginosa*) and Gram-positive bacteria (*S. aureus*, *Bacillus*) were used as experimental objects to evaluate the bactericidal efficacy and broad spectrum of the Co_3O_4 catalyst [34]. The model bacteria and marine bacteria were cultured in 2216E medium (containing 5 g/L tryptone, 1 g/L of yeast extract, 0.01 g/L ferric phosphate) and Luria-Bertani (LB)

medium (containing 10 g/L of sodium chloride, 10 g/L tryptone, 5 g/L of yeast extract), respectively. After the bacteria were cultured in a shaker with a rate of 180 rpm for 24 h, the concentration of bacteria reached $10^8\text{ CFU}/\text{mL}$. The sterilized seawater was electrolyzed by chronopotentiometry ($25\text{ mA}/\text{cm}^2$) for 10, 20, 30, and 40 mins, and then 1 mL of the electrolyte containing active chlorine was extracted, respectively. The same volume of the electrolyte containing different concentrations of active chlorine and the diluted bacterial solution were fully mixed and reacted statically for 30 mins. Then, continue to dilute the mixture to the appropriate concentration. Finally, the $50\text{ }\mu\text{L}$ mixture was put in a solid medium and shaken evenly. After 48 h, the plate count method was used to calculate the number of bacteria with three repeats.

2.7. Evaluation of antifouling performance

Fig. 2 shows the laboratory antifouling system and marine antifouling system of optical glass and the schematic diagram of the sterilization and antifouling process of $\text{Co}_3\text{O}_4/\text{TM}$ is shown in Fig. S1. The double electrode system with $\text{Co}_3\text{O}_4/\text{TM}$ as anode and TM as cathode was placed next to the glass window of an undersea searchlight. The laboratory antifouling system was immersed in a mixture of 2216E medium and natural seawater with a volume ratio of 1:9, the chronopotentiometry ($1\text{ mA}/\text{cm}^2$) was operated for 1 h per half-day, and 1/10 vol of solution was renewed every-two days. The glass slide was taken out for recording on different experimental days. The underwater transmittance of the glass sample was measured three times by a UV-vias spectrometer (Hitachi U3900). The attachment of the bacteria on the glass surface was observed through metallographic/fluorescence microscope (OLYMPUS BX53M, Japan) before (metallographic

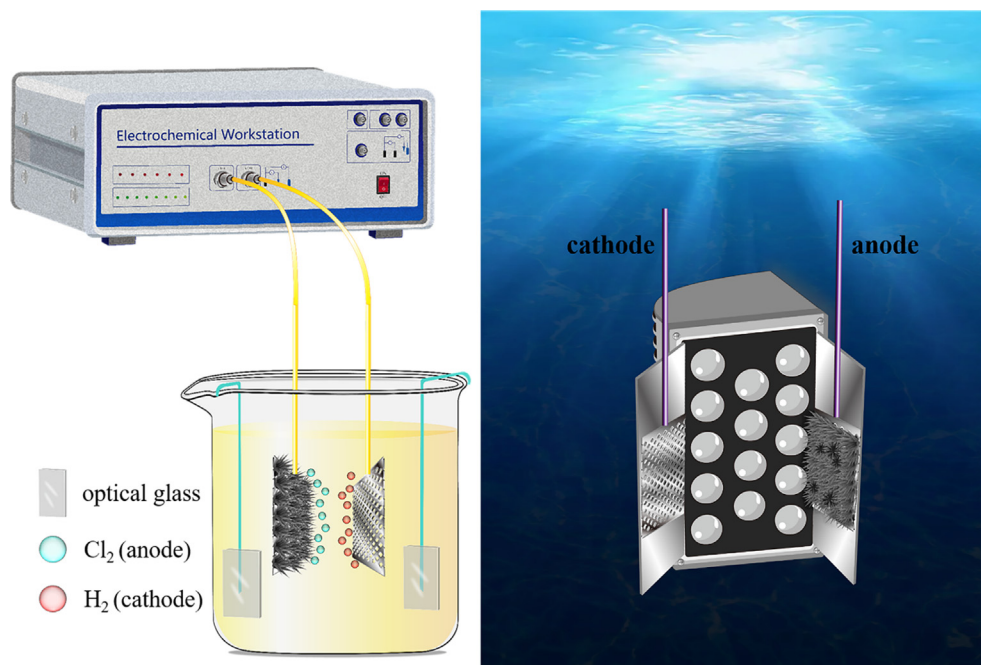


Fig. 2. The electrochemical chlorination antifouling system of optical glass in the laboratory (a) and marine (b).

picture) and after (fluorescence picture) the staining of the LIVE/DEAD (R) BacLight (TM) Bacterial Viability Kit, and the fluorescence images were counted three times. The three components (bacteria, protein, and polysaccharide) of the biofilm were stained with three kinds of dyes (DAPI, FITC, and ConA-Texas red), respectively, and then the distribution of the three components on the glass surface was observed by 3D confocal laser scanning microscope (3D-CLSM, CZ micro imaging GmbH).

3. Results and discussion

3.1. Structural and textural characterizations of the catalysts

was used to observe the morphology and structure of Co₃O₄ powder synthesized by the hydrothermal method under different conditions. Fig. 3 exhibits the low (a_1 - d_1) and high-magnification (a_2 - d_2) SEM images of the four kinds of powder after calcination, and the uncalcined samples are shown in Fig. S2, which clearly indicate that the samples with four structures have been successfully synthesized and the four kinds of powder still maintain stable morphology after calcination. As opposed to the uncalcined sample, the calcined sample has a rougher surface, which may be caused by the combustion of carbon and the removal of OH⁻ ions in the sample. The high-magnification image shows that the nanosheet samples are stacked and aggregated into a thick sheet structure. The nanorods show different lengths with a width of about 100 nm. The diameter of nanourchin is about 0.9–1.2 μ m, and that of nanoflower is about 1.7–2 μ m. Meanwhile, Fig. 3e1 and e2 show SEM images of the surface of the TM electrode prepared under the synthetic conditions of the nanourchin. The images show that the Co₃O₄ nanourchin is obtained on the surface of TM by the hydrothermal method and forms a uniform layer.

Fig. 4a shows the XRD pattern of Co₃O₄ calcined at 300 °C with four different morphologies. The peak at 31.3°, 36.8°, 38.5°, 44.8°, 55.7°, 59.4°, 65.2°, and 77.3° agrees with the (220), (311), (222), (400), (422), (511), (440), and (533) planes of the spinel Co₃O₄ phase (JCPDS Card no. 43–1003), respectively. No peaks from other phases or impurities are observed, which indicates that the cobalt precursors were completely converted into pure Co₃O₄ phase after

being calcined in air at 300 °C for 2 h. The diffraction peaks of the four kinds of powder are high and sharp, which indicates their good crystallinity. The Raman spectrum of the Co₃O₄ powder was measured at room temperature and are shown in Fig. 4b. In the spectrum, the peak at 193, 473, 516, 614, and 680 cm⁻¹ are designated as the F_{2g}, E_g, F_{2g}, F_{2g}, and A_{1g} modes of spinel Co₃O₄, respectively.

Fig. 4c and d show the N₂ adsorption–desorption isotherms and pore size distributions of the Co₃O₄ powder at –195.85 °C. The N₂ adsorption–desorption isotherms were calculated through the BET method to get a typical type IV isotherm and hysteresis loop of type H3. The results indicate that the samples contain mesopores and the pore structure is irregular [35]. Among them, Co₃O₄ with nanosheet and nanorod structure has a relatively narrow hysteresis loop near the relative pressure of 0.9 which corresponds to the characteristic of particulate materials. The voids in the materials may be caused by the self-assembly and stacking of nanoparticles which usually show wide pore distribution and moderate mesopores [36]. Fig. 4d shows the pore size distribution calculated by the BJH method. The nanosheet and nanorod samples show a broad peak structure, mainly with medium mesopores of about 22 nm. Meanwhile, the nanourchin and nanoflower samples have a sharp peak which indicates the pores have a size of around 8 nm. In addition, Fig. S3 summarizes the BET surface area, pore volume, and pore size of all samples, and their specific values are shown in Table 1. The nanourchin sample has the largest BET surface area and then is nanoflower, nanorod, and nanosheet. The pore volume has the same trend as the BET surface area, but the pore size has an opposite trend. The nanourchin and nanoflower structures have larger pore volumes with smaller pore sizes, which indicates that they have more pores than nanorod and nanosheet structures. A larger number of pores leads to more available channels, which means that they have a larger permeability of electrolyte solution, a larger surface for Cl₂ releasing, and higher Cl₂ diffusion.

3.2. Electrocatalytic properties of catalysts

To further study the relationship between their electrocatalytic properties and morphologies, the electrocatalytic properties of four

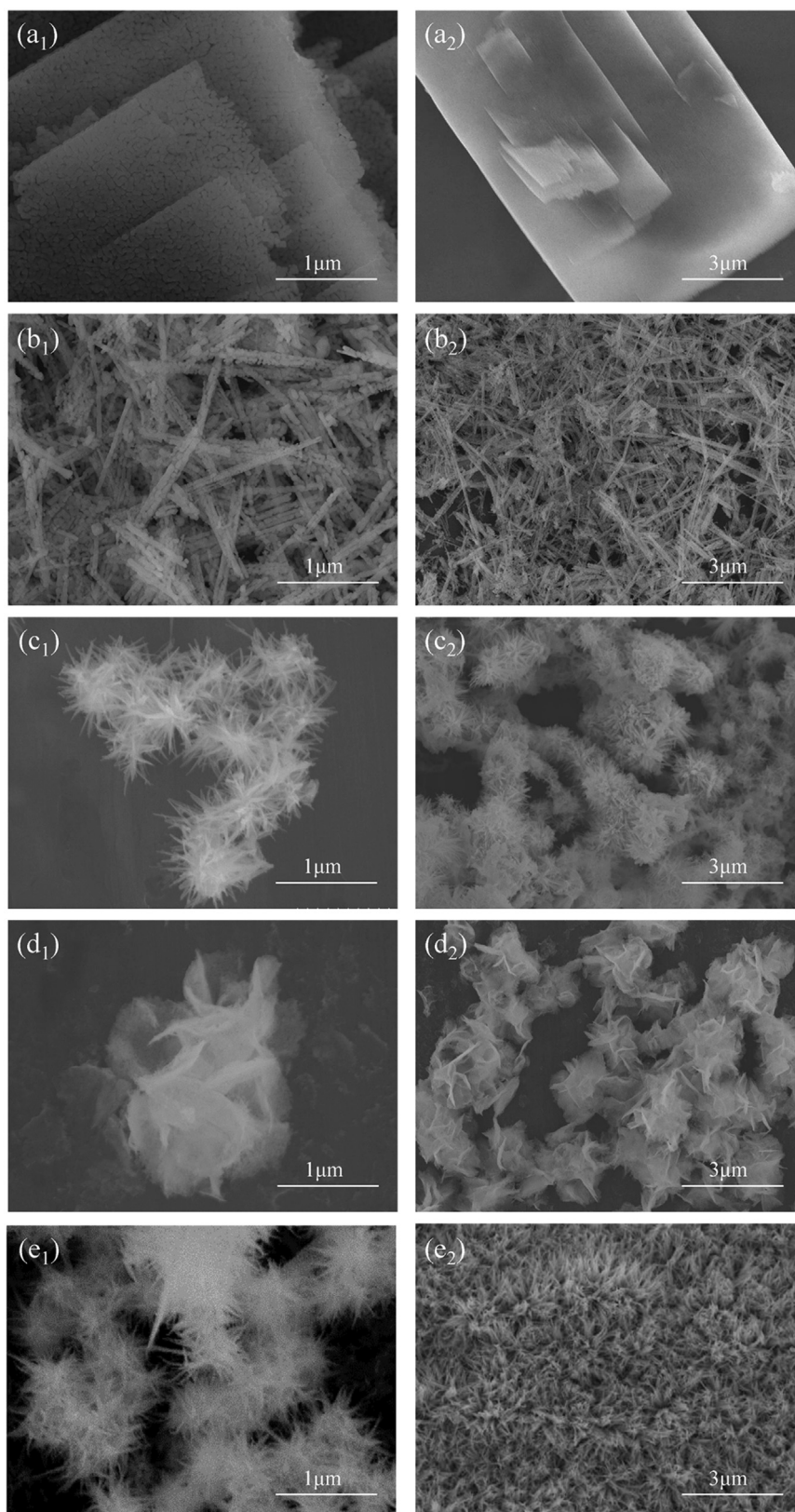


Fig. 3. SEM images of the Co_3O_4 samples: nanosheet (a_1 and a_2), nanorod (b_1 and b_2), nanourchin (c_1 and c_2), nanoflower (d_1 and d_2), and $\text{Co}_3\text{O}_4/\text{TM}$ electrodes (e_1 and e_2).

kinds of Co_3O_4 powder were analyzed by LSV, as shown in Fig. 5. Fig. 5a shows that, before the theoretical chlorine evolution potential, the current density of the samples increases with the increase of the potential. It indicates that the samples get a good CER

response. The curve of nanourchin sample shows the largest slope that indicates the nanourchin structure has the best electrocatalytic property. The slope of the Tafel diagram is one of the important parameters for studying the catalyst reaction mechanisms and

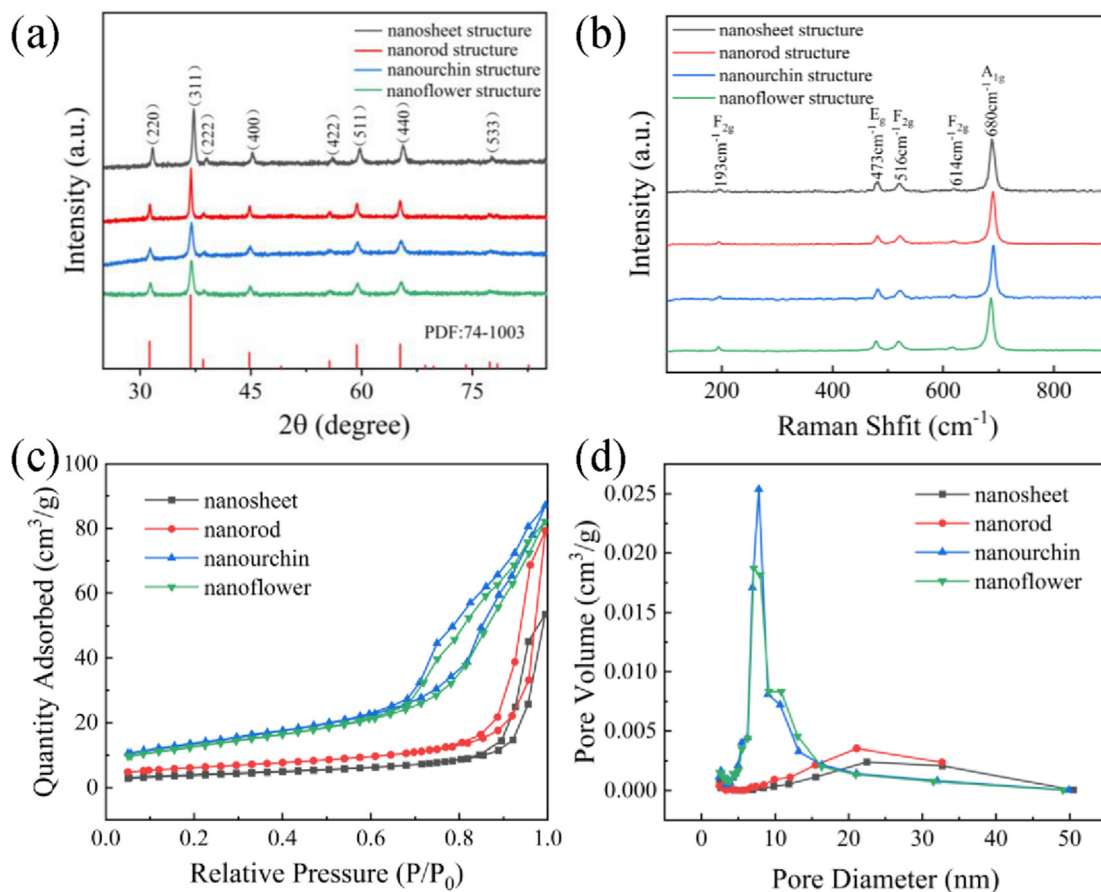


Fig. 4. XRD patterns (a), Raman spectra (b), N_2 adsorption–desorption isotherms (c), pore size distributions (d) of the Co_3O_4 electrocatalysts with different morphologies after annealing.

Table 1

The BET surface area, pore-volume, and pore size of the Co_3O_4 powder catalysts with four structures.

Structure	BET	Pore Volume	Pore Size
Nanosheet	13.9261	0.08248	25.7995
Nanorod	21.4102	0.10447	22.7637
Nanourchin	47.6527	0.13436	10.5419
Nanoflower	44.7687	0.12782	10.7744

evaluating reaction kinetics [37]. The Tafel slopes of different structures are calculated by the following Tafel formulation according to LSV (Fig. 5b):

$$\eta = a + b \log j \quad (1)$$

where η is the overpotential, a is the cathodic intercept that is related to the exchange current density, b is the Tafel slope, and j is the current density. It can be seen that the Tafel slopes of nanosheet, nanorod, nanourchin, and nanoflower structure catalysts are 223.58, 207.78, 149.58, and 176.09 mV/dec, respectively. A smaller Tafel slope indicates that the catalyst has better reaction kinetics and the required current density can be obtained at lower potential. It means that the sample with nanourchin structure has the best electrochemical activity. In addition, the corresponding overpotentials of the four kinds of powder catalysts in neutral pH solution are shown in Fig. S4. Among them, the nanourchin structure has the lowest overpotential, indicating that it has more active reaction centers. As shown in Fig. 5c, EIS measurements of Co_3O_4 powders show that the nanourchin structure has the lowest charge

transfer resistance, which means it has the fastest charge transfer rate. These results indicate that the nanourchin structure has the best electrochemical performance of CER.

The relationship between the specific surface area and the electrocatalytic performance of the materials is also evaluated by the electrochemical surface area (ECSA) as follows [38]:

$$ECSA = C_{dl} / C_s \quad (2)$$

where C_{dl} is the double-layer capacitance and C_s is the specific capacitance of the smooth sample surface under the same conditions. C_s is a fixed value for the same material so ECSA is proportional to C_{dl} here. C_{dl} can be obtained from the CV. The CV curves of the four powders are shown in Fig. S5. With a scanning potential range of 0.4–0.5 V, different scanning rates of 10–45 mV/s were selected for testing in the non-Faraday current response region. As shown in Fig. 5d, the current density is obtained by fitting CV curves with a potential of 0.45 V at different scanning speeds. The relationship between the current density and scanning rate of all four morphologies shows approximately linear behavior. For CER, the C_{dl} of nanosheet, nanorod, nanourchin, and nanoflower structure was calculated to be 5.89, 7.02, 10.23, and 8.03 mF/cm², respectively. The larger ECSA indicates that the nanourchin structure has electrocatalytic performance, which well agrees with previous results.

On account of the best electrocatalytic performance, the nanourchin Co_3O_4 was synthesized directly on the surface of TM. The sample was used as the electrode to carry out a series of experiments and the commercial DSA was used as a control due to its catalytic activity for CER. The LSV curves of Co_3O_4 /TM, DSA, and

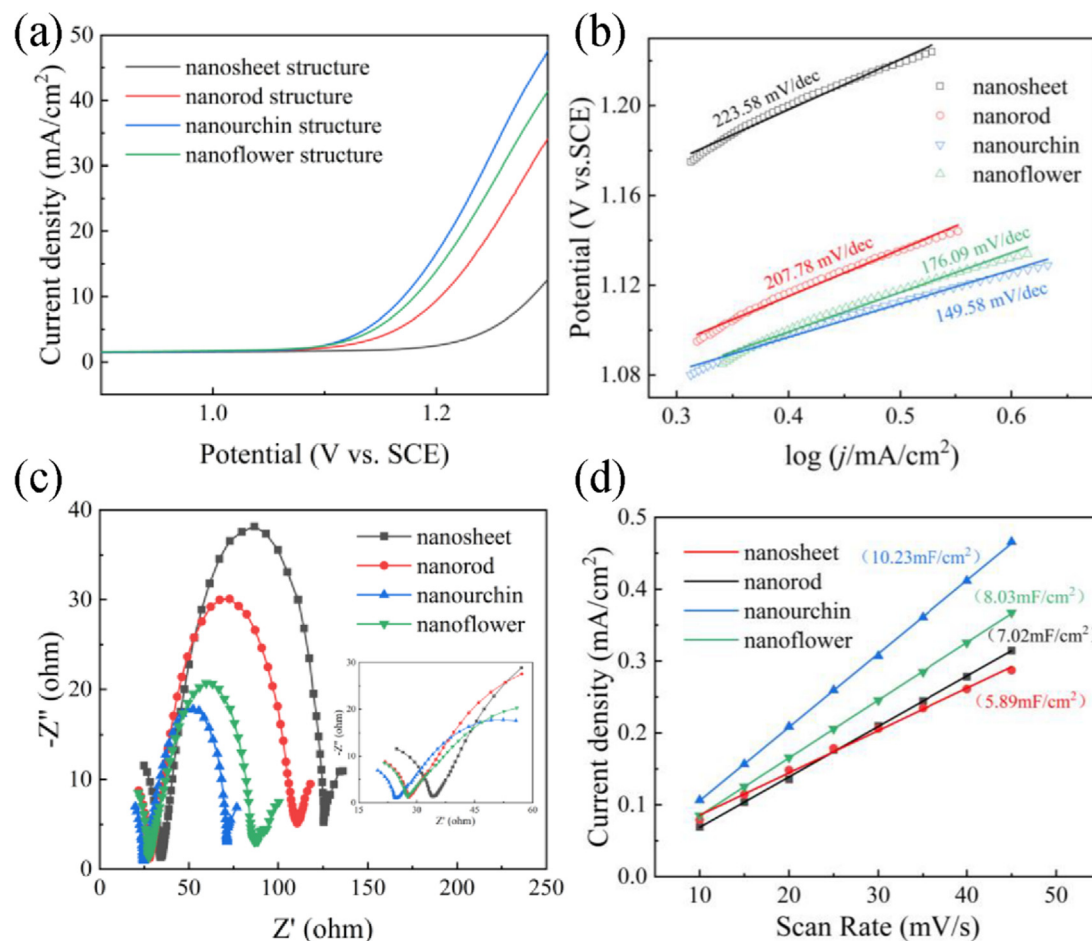


Fig. 5. LSV curves (a) and Tafel plots (b) responses of different morphologies samples with glassy carbon electrodes in 0.6 M NaCl solution. (c) EIS measurements of the Co_3O_4 anodes with different morphologies. (d) The linear slope equal to twice that of C_{dl} (calculated by the CV curve at 0.45 V) is used to represent ECSA of different morphologies.

bare TM in 0.6 M NaCl solution and seawater are shown in Fig. 6a, respectively. It shows that the electrocatalytic activity of $\text{Co}_3\text{O}_4/\text{TM}$ is much higher than that of DSA, and bare TM in both electrolytes. Meanwhile, compared with the current density in the NaCl solution, that of $\text{Co}_3\text{O}_4/\text{TM}$ in seawater maintains 84.2 % which is only

74.6 % for DSA. The results conclude that the $\text{Co}_3\text{O}_4/\text{TM}$ electrode has better activity and adaptability than DSA in the marine environment. To verify the CER selectivity of $\text{Co}_3\text{O}_4/\text{TM}$, LSV curves were measured in solutions with and without chlorine. In Fig. 6a, the electrochemical performance of $\text{Co}_3\text{O}_4/\text{TM}$ in 0.6 M NaCl is

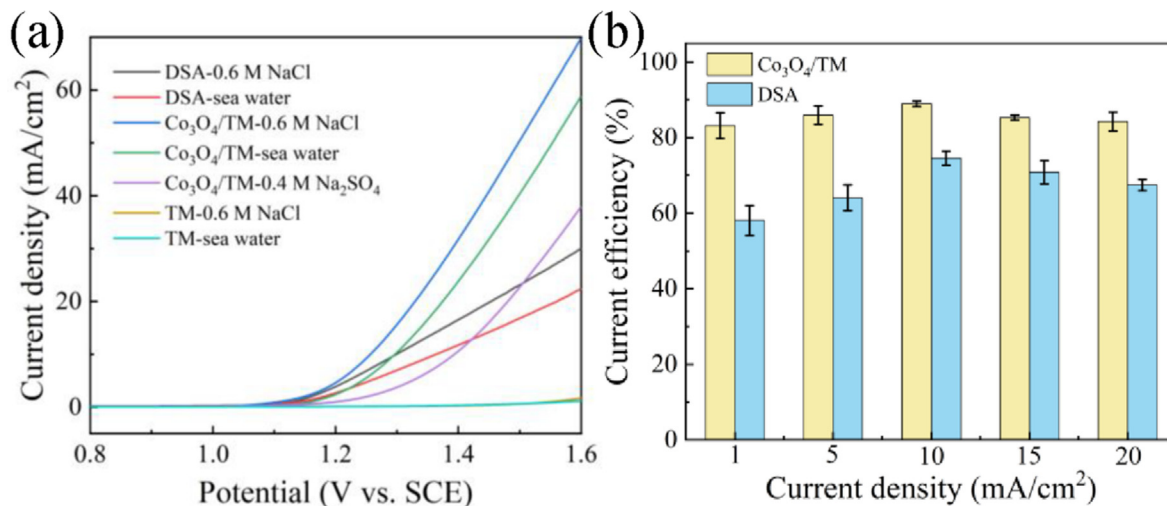


Fig. 6. (a) LSV curves responses of the $\text{Co}_3\text{O}_4/\text{TM}$, DSA, and bare TM anodes in different solutions. (b) Current efficiency for CER of $\text{Co}_3\text{O}_4/\text{TM}$ and DSA electrodes at different current densities.

higher than that in 0.4 M Na₂SO₄. And the onset potential of Co₃O₄/TM in 0.4 M Na₂SO₄ is higher than that in 0.6 M NaCl. The prepared Co₃O₄/TM electrode only carries out CER at low potential, and simultaneously carries out CER and oxygen evolution reaction (OER) at high potential. These results mean that the electrocatalyst has higher selectivity for CER. To further study the ratio of OER to CER at different current densities, the current efficiency is measured by DPD colorimetry which is a common way to compare the selectivity of CER [39,40]. The corresponding current efficiency is calculated by Eq. (3).

$$\mu = \frac{2VF [Cl_{2(DPD)}]}{jAt} \quad (3)$$

where ε is the current efficiency, F is the Faraday constant (96487C/mol), V is the volume of the electrolyte, $[Cl_{2(DPD)}]$ is the molar concentration of the active chlorine, A is the area of the electrode, j is the applied current density (1, 5, 10, 15, 20 mA/cm²), and t is the electrolysis time (180 s). Fig. 6b shows the current efficiency of Co₃O₄/TM and DSA for CER at different current densities in 0.6 M NaCl solution. Hence, the current efficiency of Co₃O₄/TM is higher than that of DSA at every current density and also higher than 80 % in our current density range. It means that the CER ratio of Co₃O₄/TM at different current densities is much higher than OER, which proves that Co₃O₄/TM has excellent CER selectivity and is better than DSA. It is worth mentioning that when the current density increases gradually, the current efficiency of Co₃O₄/TM and

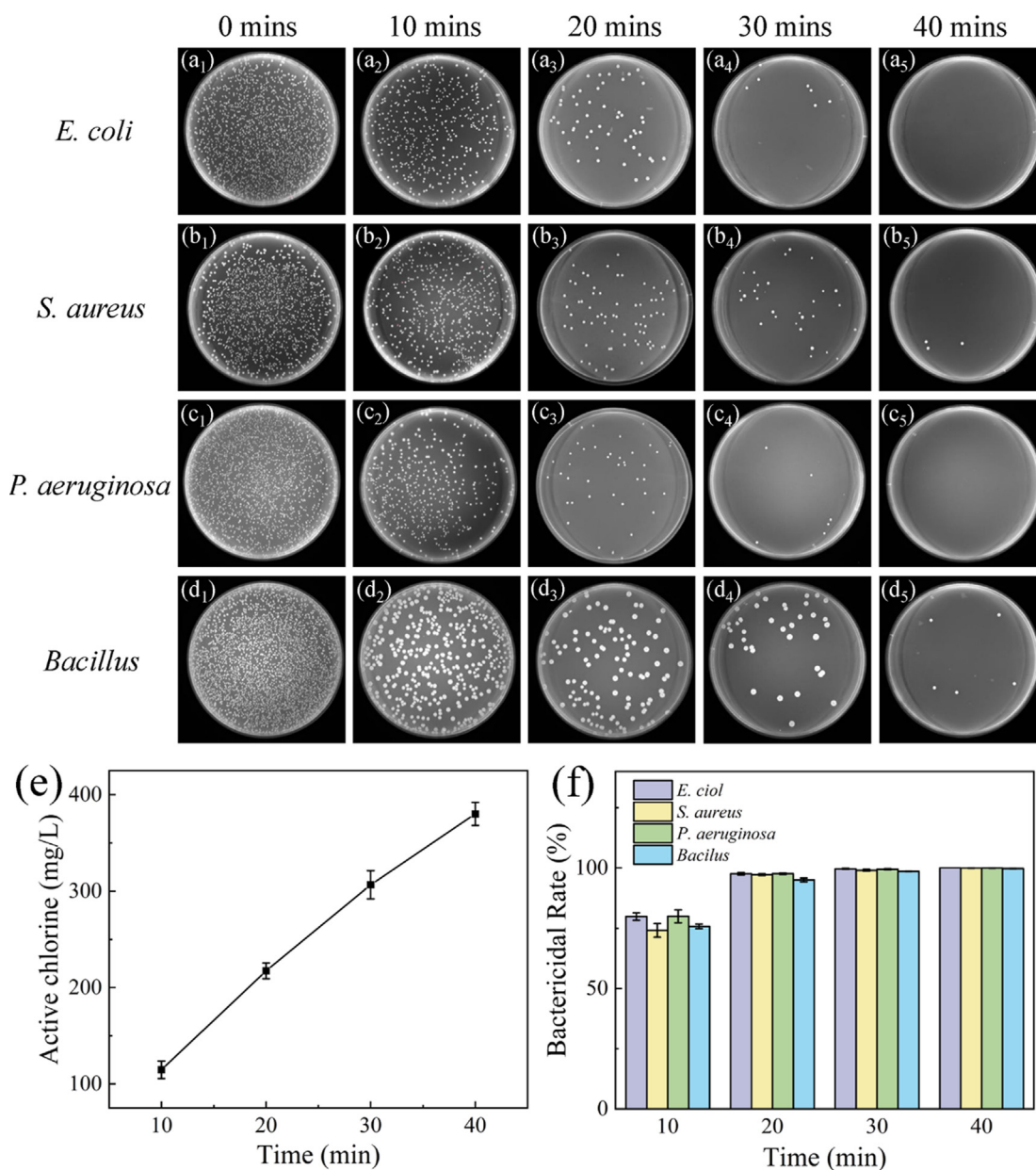


Fig. 7. *E. coli* (a₁-a₅), *S. aureus* (b₁-b₅), *P. aeruginosa* (c₁-c₅) and *Bacillus* (d₁-d₅) treated with different concentrations of active chlorine (Co₃O₄/TM electrode chlorinated at 25 mA/cm² current density by chronopotentiometry for 0, 10, 20, 30 and 40 mins, respectively). (e) The concentration of active chlorine under different electrolysis times. (f) Bactericidal rate of four kinds of bacteria treated with the electrolyte of different electrolysis time (40 mins corresponds to a sterilization rate of 100 % without error bars).

DSA increases at first and then decreases. This phenomenon can be attributed to the competition between OER and CER at the high anodic potential [41–43]. Fig. S6 shows the catalytic stability of $\text{Co}_3\text{O}_4/\text{TM}$ and DSA electrodes with a current density of 5 and 10 mA/cm^2 , respectively. Both $\text{Co}_3\text{O}_4/\text{TM}$ catalyst and DSA sample show a relatively constant potential in 0.6 M NaCl solution within 10 h. It means that the $\text{Co}_3\text{O}_4/\text{TM}$ electrode has the same excellent stability as the commercial DSA electrode.

3.3. Bactericidal analysis of catalysts

As we all know, chlorine disinfection mainly depends on its reaction with water to generate HOCl and OCl^- . Because HOCl has a small molecular weight and is neutral, it can penetrate through the cell wall of bacteria. The strong oxidation of OCl^- kills bacteria thoroughly to achieve the purpose of sterilization [44,45]. In this paper, Gram-negative bacteria (*E. coli*, *P. aeruginosa*) and Gram-positive bacteria (*S. aureus*, *Bacillus*) were used as experimental objects to evaluate the bactericidal ability of the Co_3O_4 catalyst

and the performance was illustrated through the plate counting method. Fig. 7a to d show that the germicidal effect of the Co_3O_4 catalyst on all four bacteria is more and more significant with the extension of electrolysis time, and almost no colonies grow on the plate after electrolysis for 40 mins. It indicates that the Co_3O_4 catalyst has very good bactericidal efficiency and broad-spectrum activity. Fig. 7e shows the relationship between electrolysis time and available chlorine concentration. The available chlorine concentration reaches 217 mg/mL and 380 mg/mL after 20 mins and 40 mins of electrolysis, respectively. The relationship between electrolysis time and the germicidal rate has shown in Fig. 7f. The bactericidal rate of model bacteria and fouling bacteria is larger than 75 % after 10 mins and reaches 95 % after 20 mins. It indicates that a chlorine concentration of 217 mg/mL can effectively kill bacteria. The bactericidal rate is almost 100 % after electrolysis for 40 mins. In addition, the germicidal efficacy of active chlorine on mixed bacteria in seawater was verified through fluorescence staining. As shown in Fig. S7, in the bacterial solution without active chlorine, almost all the bacteria can keep alive.

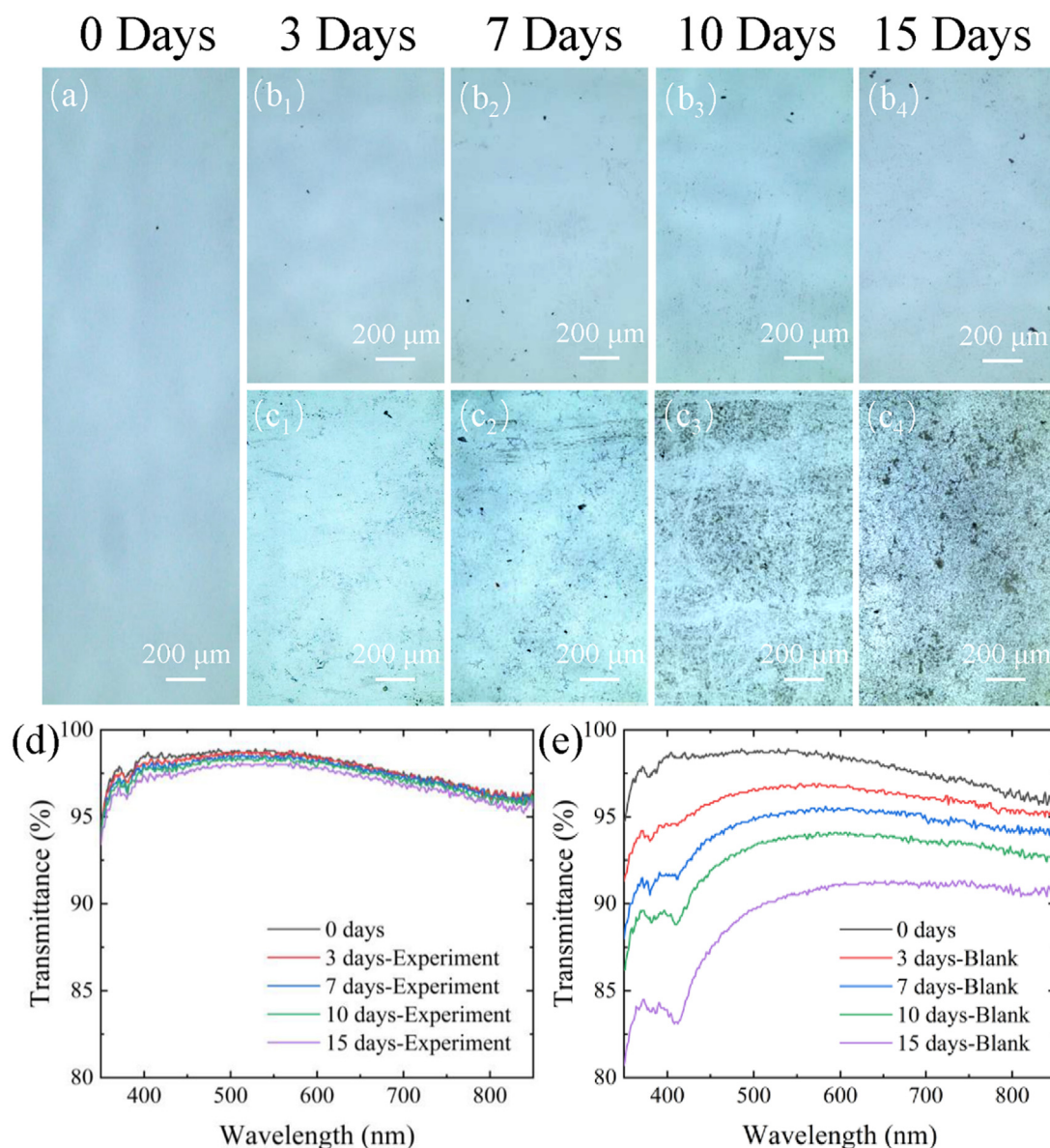


Fig. 8. Microscope images of bare (a), experimental group (b_1 – b_4), and blank group (c_1 – c_4) optical glass for different days (1–4: immersion times of 3, 7, 10, and 15 days, respectively). Underwater transmittance spectra of the experimental group (d) and blank group (e) for different days in seawater with bacteria.

However, with a low concentration of active chlorine, only a few bacteria can survive. Finally, with a high concentration of active chlorine, all the bacteria in the solution are killed. It further confirmed that the active chlorine prepared by $\text{Co}_3\text{O}_4/\text{TM}$ electrode has good bactericidal broad-spectrum activity.

3.4. Antifouling performance of catalysts

The antifouling performance of chlorine produced by the $\text{Co}_3\text{O}_4/\text{TM}$ electrode was tested through laboratory experiments, as shown in Fig. 2a. In the experiments, the double electrode system, with $\text{Co}_3\text{O}_4/\text{TM}$ as the anode, and TM as the cathode, was placed next to a glass slide in seawater containing mixed bacteria. The current density of 1 mA/cm^2 was applied to the anode ($\text{Co}_3\text{O}_4/\text{TM}$) and cathode (TM) for 1 h, and the electrified chlorination was carried out every 12 h. Fig. 8a, b, and c show the microscopic

images of glass surfaces with different soaking days. With the increase of soaking days, compared to the dirty control slide without electrodes, the slide with electrodes remains almost clean and transparent even after soaking for 15 days. The underwater-transmittance spectra of glass before and after chlorine protection by the electrochemical method under different immersion days showed in Fig. 8d and e. Similar to the results of microscopic images, the transmittance of the blank control group decreases significantly with the increase of soaking days due to the continuous accumulation of biological dirt on the surface. In contrast, the light transmittance of the experimental group keeps almost the same as that of the clean glass. As shown in Fig. S8, the line chart of the transmittance of the glass in the experimental group and the blank group at 550 nm, more intuitively reflects the changes in transmittance. After soaking in bacterial solution for 15 days, the transmittance of the glass in the experimental group decreases by less than

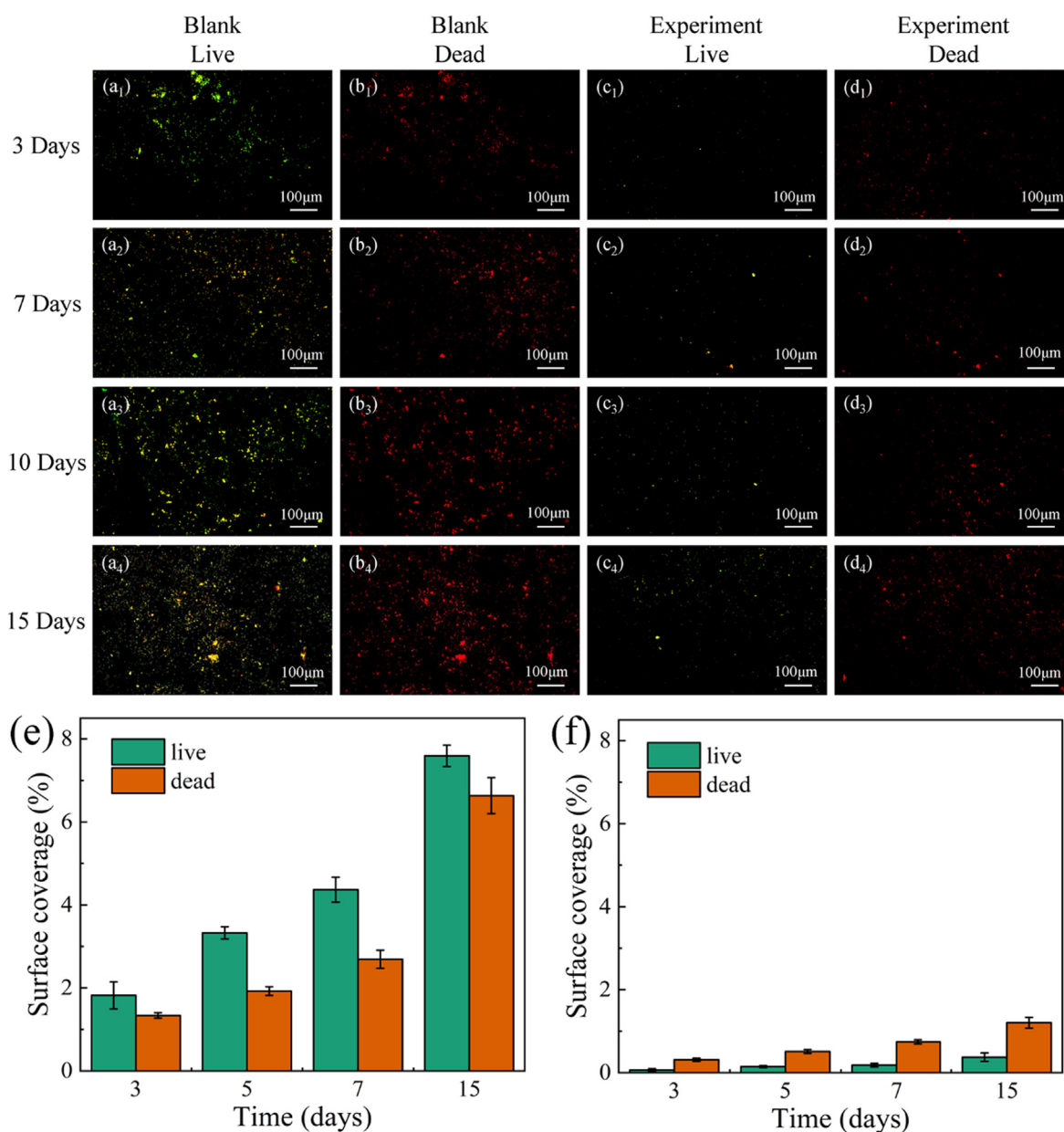


Fig. 9. Fluorescence microscope images of optical glass for different days in seawater with bacteria (a, b: blank group; c, d: experimental group; green, red: live bacteria, dead bacteria; 1–4: immersion time of 3, 7, 10 and 15 days, respectively). Surface average of live and dead bacteria on blank group optical glass (e) and experimental group optical glass (f) on different days.

1 %, while that in the blank group decreases by nearly 10 %. The results indicate that there was no biological fouling on the protected glass surface. Therefore, the Co_3O_4 catalyst has a significant effect on antifouling and protecting the surface transmittance of the optical window in the environment of seawater.

To further verify the Sterilization and antifouling performance of the $\text{Co}_3\text{O}_4/\text{TM}$ electrode, the adhesion of bacteria on the glass surface was measured by the fluorescent live/dead stain method. The variation of bacteria adhesion on the glass surface of the experimental group and blank group with time is shown in Fig. 9a to d. Fig. 9e and f provide the settlement density and live/dead state of bacteria on the glass surface (red and green represent dead bacteria and live bacteria, respectively). The fluorescence images show that the adhesion number of bacteria on the glass surface of the blank control group increases with the increase of soaking days. Even if the glass was only soaked for 3 days, the bacteria can easily settle on the glass surface. According to the data on bacterial settlement rate, the average coverage rate of live/dead bacteria in the blank group is 1.82 %/1.33 % and 7.59 %/6.63 % in 3 days and 15 days, respectively. In contrast, the total bacterial coverage of the glass in the experimental group is 0.37 %/1.20 % after soaking for 15 days. Intriguingly, the coverage rate of living bacteria on the blank sample is higher than that of dead bacteria, while it is lower on the experimental sample. This can be attributed to the killing effect of HOCl on bacteria and the natural death of bacteria by metabolism. Consequently, the method of producing chlorine by $\text{Co}_3\text{O}_4/\text{TM}$ electrode can effectively inhibit the settlement of bacteria on the surface of the optical window.

To study the antifouling principle of the Co_3O_4 catalyst electrode, the morphology of biofilms on the surface of optical window in seawater was detected by 3D-CLSM [46]. The adhesion of bacteria depends on adhesin which is mainly glycoprotein. When the optical glass is placed in the seawater environment, glycoproteins and other components in the seawater will quickly attach to the glass surface to form a conditioning film, thus providing attachment points for bacteria [47,48]. In Fig. 10, CLSM images

show the 3D distribution of bacteria (blue), proteins (green), and polysaccharides (red) in the experimental group and the blank group under different soaking days. No matter 7 days or 15 days, with the effect of chlorine, the bacterial adhesion rate on the glass surface of the experimental group is much lower than that of the blank group. At the same time, the experimental data show that the contents of polysaccharides and proteins on the glass surface of the experimental group are significantly lower than those of the blank group. The results show that the formation of conditioning film can be inhibited under the action of active chlorine, thus the attachment sites of small bacteria can be greatly reduced. When the attachment of small bacteria is inhibited, the number of attachment sites of some large microorganisms will also decrease, thus further preventing the formation of biofouling on the surface.

4. Conclusion

In conclusion, to better understand the active chlorine antifouling performance of Co_3O_4 electrocatalysts, four nano samples with different microscopic morphologies, nanosheet, nanorod, nanourchin, and nanoflower, were prepared through the hydrothermal method. The electrocatalytic tests show that nanourchin Co_3O_4 has the best performance for chlorine evolution due to its maximum Brunauer-Emmett-Teller and electrochemical surface area. Therefore, to continually explore its antifouling performance, a nanourchin Co_3O_4 coated titanium mesh electrode was developed for the antifouling study. Electrochemical tests show that Co_3O_4 electrode has better catalytic performance and selectivity than the commercial dimensionally stable electrode. The antifouling experiments show that the electrochemical chlorination system with Co_3O_4 electrode can effectively inhibit the adhesion of microorganisms. All these results indicate that Co_3O_4 electrocatalyst has a good application prospect for the antifouling protection of optical windows in the ocean.

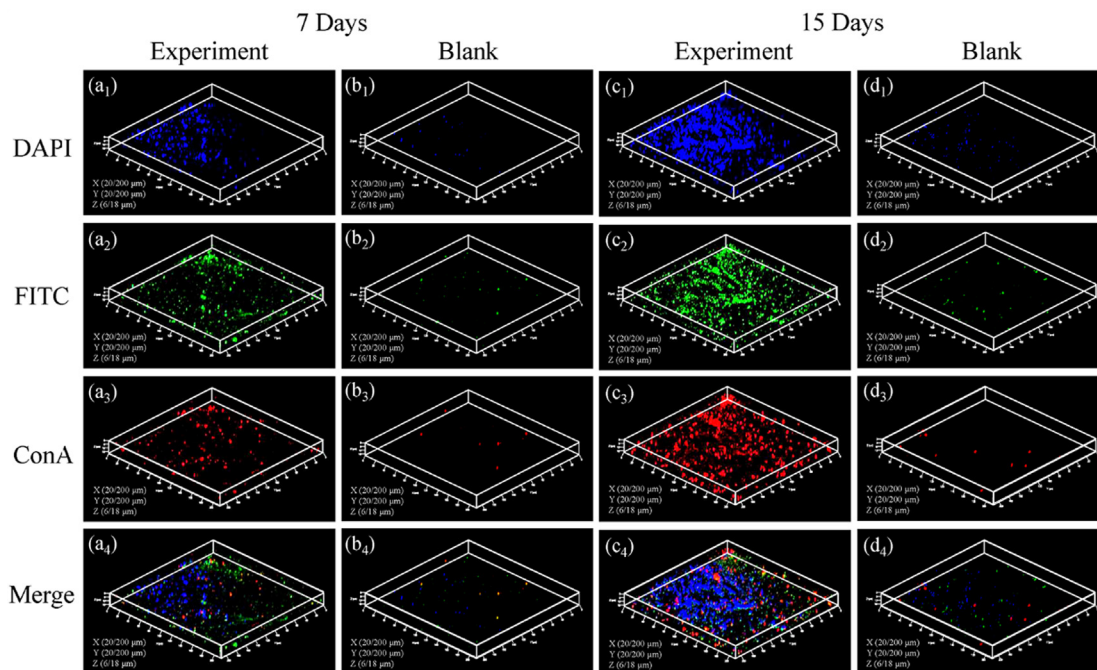


Fig. 10. Confocal laser scanning microscopy images of blank group optical glass samples (a and c represent 7 and 15 days, respectively) and experimental group optical glass (b and d represent 7 and 15 days, respectively) for different days in seawater with bacteria. In each case, 1, 2, 3, and 4 indicate DAPI, FITC, ConA-Texas, and Merge, respectively.

CRediT authorship contribution statement

Haomin Wang: Formal analysis, Investigation, Data curation, Writing – original draft. **Jiang Li:** Writing – review & editing. **Dun Zhang:** Funding acquisition. **Peng Wang:** Conceptualization, Methodology, Funding acquisition. **Guoqing Wang:** Conceptualization, Methodology, Supervision.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2022.08.004>.

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